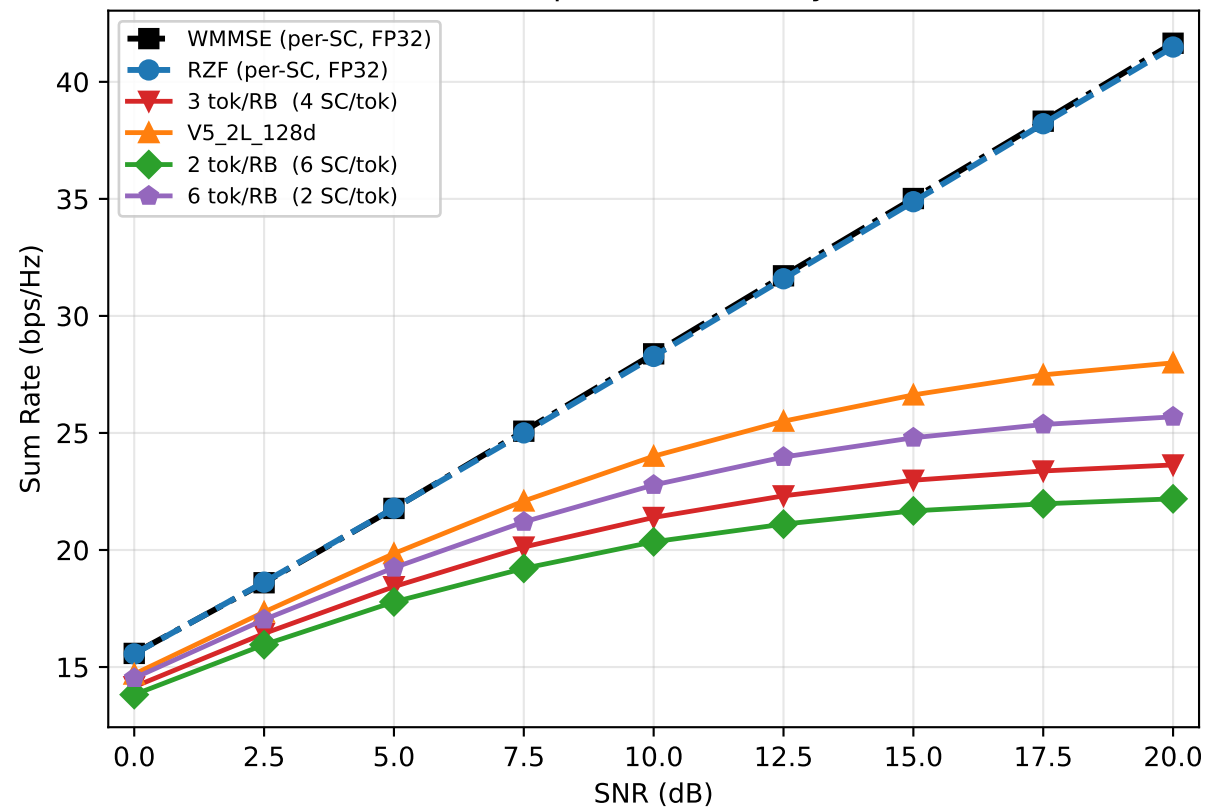
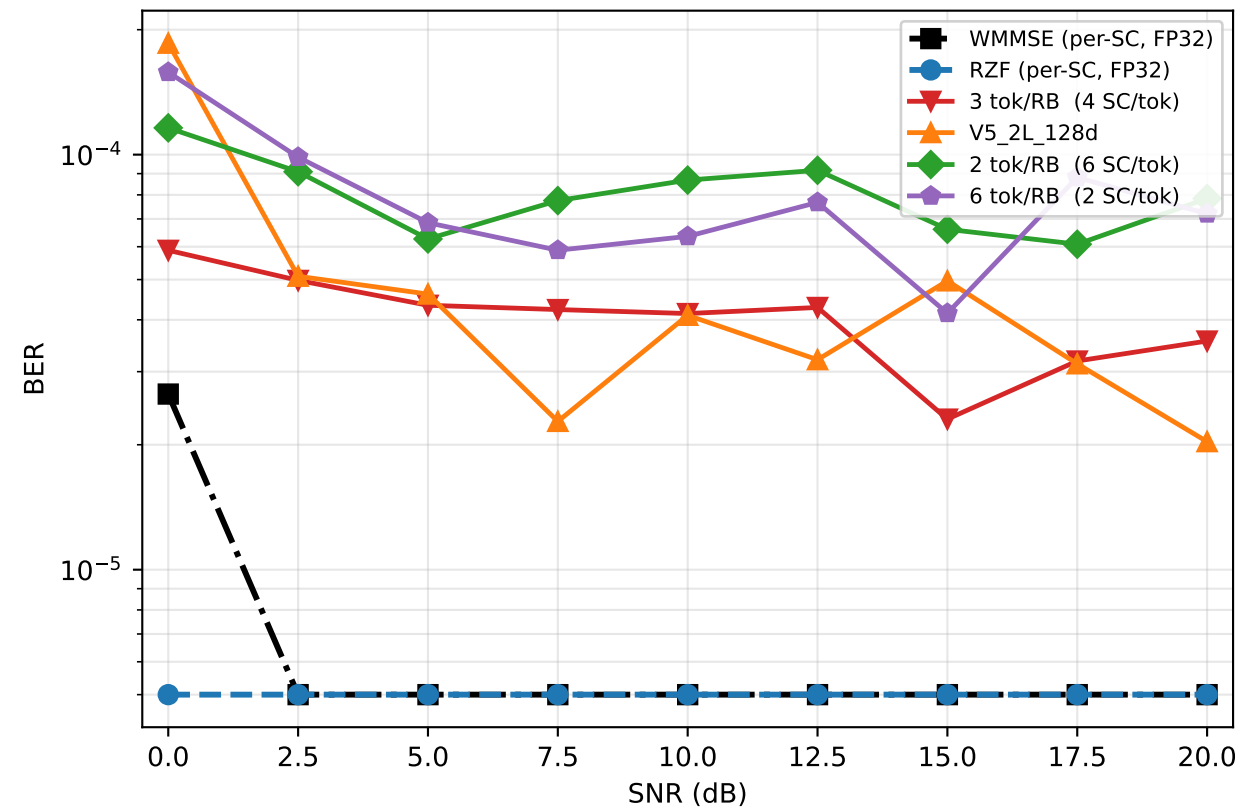


V4.2 Transformer Precoder — 8×4 MU-MIMO (8 TX, 4 UE, 72 SC, SNR train = 15 dB)
Ablation study: tokens per Resource Block (RB = 12 SC)

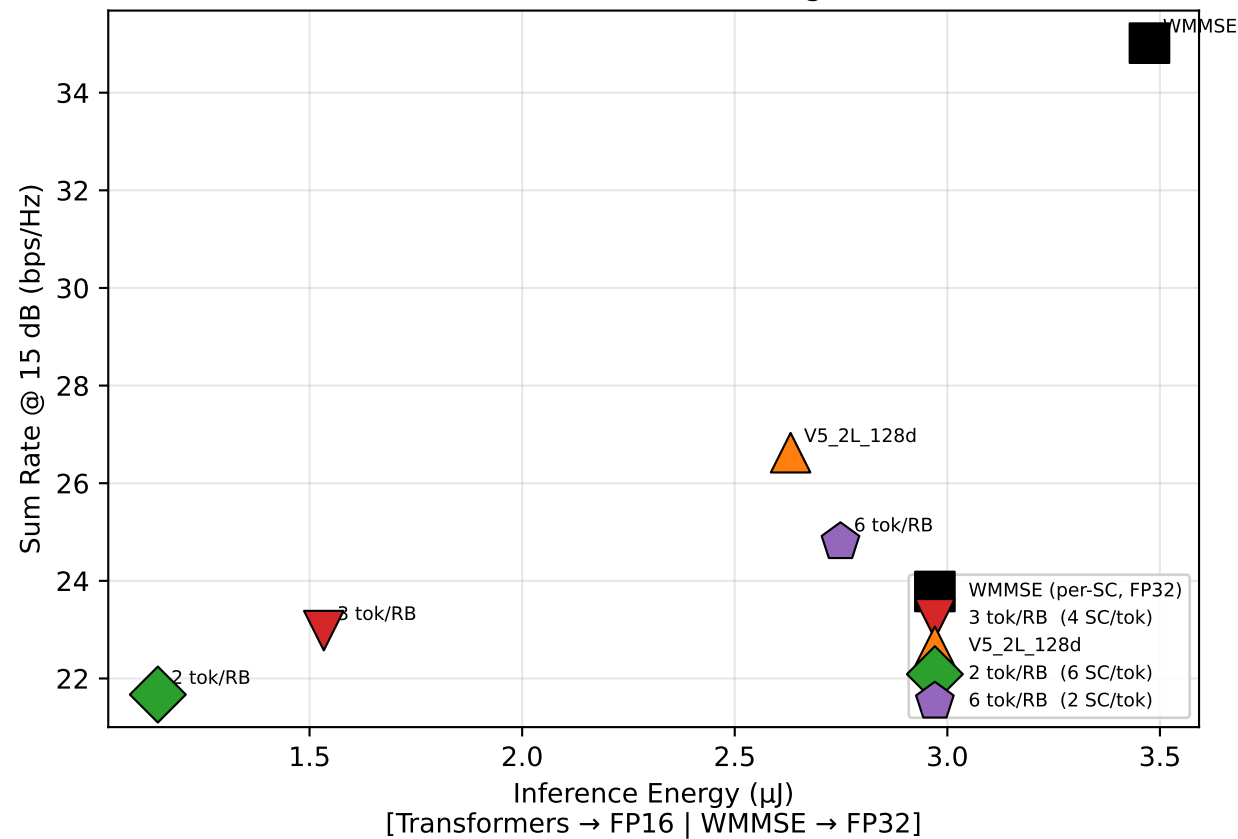
(a) Spectral Efficiency



(b) Bit Error Rate (Monte Carlo accumulation)



(c) Energy-Performance Pareto
(RZF excluded — see Fig. 2)



(d) Pareto Summary @ 15 dB
 Energy gain = WMMSE_FP32 / model_FP16

Method	Rate (bps/Hz)	Gap vs WMMSE	FLOPs (M)	Energy FP16 (μ J)	Gain vs WMMSE (FP32)
WMMSE (per-SC, FP32)	35.02	+0.00	1842	N/A	ref
RZF (per-SC, FP32)	34.88	-0.14	2	N/A	ref
V5_2L_128d	26.63	-8.39	4853	2.6314	1.3×
6 tok/RB (2 SC/tok)	24.79	-10.22	5081	2.7487	1.3×
3 tok/RB (4 SC/tok)	22.98	-12.03	2834	1.5335	2.3×
2 tok/RB (6 SC/tok)	21.67	-13.34	2112	1.1432	3.0×